

How a Scientific Consensus Collapsed

The curious case of social psychology.

By Jacob Mikanowski | February 20, 2025

Fifteen years ago, social psychology was in its pomp. It was the hottest thing in the social sciences, ubiquitous in TED Talks and on best-seller lists. Tiny, targeted, and experimentally tested changes to behavior seemed to carry a utopian promise of making life better for people as well as improving society as a whole. These “nudges” and “life hacks” found their way out of the classroom and conference hall and into the corporate boardroom and the White House. The TED Talk by Amy Cuddy, a former professor at Harvard Business School, on “power poses” racked up 74 million views on YouTube. President Obama told *Vanity Fair* that he only wore gray and blue suits to cut down on decision fatigue, an idea that was itself an outgrowth of the research into ego depletion. Cass R. Sunstein, director of the behavioral economics and public-policy program at Harvard Law, was serving as Obama’s head of information and regulatory affairs, bringing his ideas about decision architecture to the heart of government.

Fast-forward to the present, and social psychology appears to be a field in crisis. Classic studies won’t replicate. Academic superstars keep being accused of research fraud. Things have gotten bad enough that some have begun asking whether psychology is a science at all, in the Kuhnian sense of a coherent research program amassing a cumulative body of knowledge, or just a scattered bunch of research subjects in search of a paradigm.

Ruth Leys, a historian of science at the Johns Hopkins University who has spent her career writing about the development of psychiatry, psychology, and the associated sciences of the mind, belongs to this group of doubters. Her new book, *Anatomy of a Train Wreck: The Rise and Fall of Priming Research*, chronicles the meteoric ascent and swift collapse of one particular subfield of social psychology. Leys sees priming research — the study of how subtle cues can subliminally shape our thoughts and behaviors — as a microcosm of what has gone wrong in psychological research in the past generation. However, unlike many of those who have weighed in on psychology’s post-replication crisis reckoning, she doesn’t focus on experimental design or malfeasance. Rather, for Leys, the problem with the current state of psychology is not bad methods but bad ideas. Chief among these is the dangerous notion of automaticity, or the idea that most of human behavior isn’t intentionally willed but derived from unconscious responses to stimuli.

The idea that human beings are little more than living automatons is an old one, going back at least to the philosophe Julien Offray de La Mettrie, who scandalized 18th-century France with his book *Man a Machine* and later put his belief in hedonism to the ultimate test by gorging himself to death on pheasant pâté. None of Leys’s protagonists go quite that far in putting their theories into practice. They do, however, succeed in imposing their “abstract, unrealistic, and impoverished picture of actual persons” on a wide swath of research psychology. *Anatomy of a Train Wreck* is a subtle (sometimes too subtle) polemic against this two-dimensional conception of the human mind, as well as an argument for moving social psychology out of the laboratory and into the street.

Leys’s story begins in the early 1970s with the rise of attribution research,

which tried to sort the causes of human actions into “internal” or “external” factors, and, in the process, demonstrate that people were poor judges of the sources of their own behavior. This effort culminated in an influential 1977 paper by Richard E. Nisbett and Timothy DeCamp Wilson titled “Telling More Than We Can Know,” in which the authors argue that people have no access to the reasons behind their own decision-making and, indeed, may not be “intentional creatures” at all.

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Every discipline speaks its own language and has its own standards of proof. Those in social psychology appear — at least to this reader — to be remarkably permissive. One of the great surprises of *Anatomy of a Train Wreck* is the gulf between the vast inferences about human nature that researchers make and the very modest experiments that sustain them. For instance, one of the key pieces of evidence in “Telling More Than We Can Know” comes from something called the stockings experiment.

In this experiment, shoppers in a store were presented with four pairs of nylon stockings laid out in a row and were asked to rate which pair seemed the best and why. By a factor of four to one, the participants said the rightmost pair was best, even though all the stockings were identical. In the postexperiment debrief, however, they weren’t able to explain *why* they chose the rightmost stockings, and denied that the stockings’ position had anything to do with their choice. An interesting finding, but it seems quite a leap to go from it to arguing that human behavior is driven by “causes external to the person as an intentional agent,” meaning that when it comes to consciousness, we’re not driving the bus: Our environment is.

This gap between evidence and claim is likewise present in the study that is at the heart of Leys’s book. In 1996, the Yale psychologist John Bargh and his collaborators Mark Chen and Lara Burrows performed an experiment on a group of 30 psychology undergraduates at New York University. The experiment had two phases. In the first, the students were given a series of scrambled phrases, which they had to arrange into grammatical sentences by eliminating one word that didn’t belong. Half the students had sentences with “neutral” words, such as *thirsty*, *clean*, or *private*. The other half featured words designed to play on stereotypes about the elderly, among them *worried*, *Florida*, *old*, *forgetful*, *lonely*, *bald*, *gray*, *wrinkle*.

In the second phase, the students were given a short debriefing by one of the experimenters on what they thought had been a language-proficiency exam. Then they walked down the corridor to the elevator and went about the rest of their day. This, unbeknownst to them, was the real test. Another experimenter, disguised as a student waiting to talk to the professor, timed how fast the students walked down the hallway. Bargh et al. reported that the cohort which had been given the elder-themed test walked considerably slower than the “neutral” cohort.

Much hung on this result. Bargh used it to argue that the students in the experiment had been “primed” by the unconscious suggestion of the elderly and

that this suggestion affected their motor function. Extrapolating outward, Bargh argued that this is how most human behavior worked. Most of the time we are responding to stimuli that we aren't aware of. Consciousness only operates after the fact, to provide rationalizations for actions we aren't responsible for.

A few freshmen taking their time going down the hallway seems like strange proof that humans are essentially somnambulists, bumbling through their lives on autopilot and using their big brains to offer the occasional excuse. But Bargh's elder experiment instantly struck a chord in the field. Daniel Kahneman would hail it as an "instant classic" in *Thinking, Fast and Slow*, and a raft of studies testing the idea of "priming" followed in its wake. Researchers tested a galaxy of stereotypes and physical stimuli and found that each had its own unique effect on behavior.

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Subjects primed with the stereotype of politicians as long-winded wrote longer essays about themselves than those who hadn't. People primed with the idea of "professor" (professor here meant to signal intelligence) did better at Trivial Pursuit than those primed with the word "hooligan." Priming with words related to assertiveness made people interrupt conversations more quickly than those primed for politeness. Unconscious thoughts related to libraries made people act quieter. Smelling cleaning fluids primed cleaning goals. Just seeing an American flag made voters more conservative. Holding a warm cup of coffee prompted a warmer evaluation of a stranger's personality.

The empire of priming began to totter in 2012, when a team of Belgian researchers headed by Stéphane Doyen reported that they were unable to replicate Bargh's elder-themed experiment. Later that same year, Kahneman spoke out, addressing an open letter in *Nature* warning that there was a "train wreck looming" in the field of priming research. In fact, the train wreck would come to encompass far more than just priming. Over the following years, dozens more studies in experimental psychology proved impossible to replicate, setting off a crisis of faith across the field.

Leys doesn't delve into the consequences of the replication crisis, however. Instead, she focuses a microscopic lens on Bargh and his critics, tallying their every parry and thrust after the initial debunking. Her focus throughout is exclusively on the idea of priming and its philosophical implications for the understanding of intention — essentially, the question of whether or not human beings have agency, in the fullest sense of the term. This approach leads her down some intriguing rabbit holes. We learn, for instance, in one amusing aside, that G.E.M. Anscombe, the 20th century's greatest philosopher of intentionality, consulted a hypnotist to quit smoking in the midst of writing her masterpiece *Intention*, and then immediately suffered a nervous attack that led her to resume smoking on the advice of her friend Ludwig Wittgenstein.

It's a shame, however, that Leys doesn't widen her gaze and place priming research in its larger cultural context, for in delving so deep into what is ultimately

a rather-small arena of psychological research, she has put her finger on a much broader trend. The two-dimensional picture of humanity buffeted by external stimuli and barely in charge of their own minds has spread from psychology to the culture at large. It shows up often in our political discourse, in talk of voters misled by propaganda or driven solely by amorphous “rage.” Social-media companies have made billions (trillions?) exploiting our fragile wills and easily diverted attentions.

And meanwhile, as psychology has lost its place in the culture, what has taken its place? Astrology, mostly. As the world of power poses and life hacks have ebbed away, zodiac signs and natal charts have surged. Paid astrology apps have gained millions of users in recent years (Co-Star alone is reported to have 30 million), and astrology influencers have amassed huge audiences on social media. In my experience, the scope of these new readings goes far beyond the newspaper horoscopes of old; suddenly, my feeds have become littered with planets in opposition and conjunction. A younger friend recently told me that they were hopeful about the next few years because the planets are back to where they were during the French Revolution.

As much as I sympathize with Leys’s call to rebuild psychology around human subjects who are conscious, intentional, and in control, I fear that there is more truth in the idea of “automaticity” than we would like to admit. At least for the time being, we are still sleepwalking, forever at the mercy of our screens and the stars.